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Linear mechanism of excitation of temperature oscillations in a metal under anomalous skin effect

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The amplitude of a temperature wave excited in a metal single crystal by an electromagnetic wave incident on its surface under the conditions of anomalous skin effect is calculated. Formulas for the normal skin effect are presented for comparison. It is shown that, for an isothermal boundary, the temperature distribution does not depend on the ratio l/δ_E (l is the electron mean free path and δ_E is the skin depth), i.e., it coincides with the temperature distribution for the normal skin effect. © 1995 American Institute of Physics.

1. The possibility of exciting temperature oscillations in a metal single crystal was predicted in Ref. 1. For this purpose, the symmetry axis of the crystal must be inclined at a certain angle φ to the normal to the sample surface that is exposed to an electromagnetic wave ($\varphi \neq 0, \pi/2$). The temperature oscillations are induced by the thermo-emf. It was shown in recent publications^{2–4} that, owing to the thermoelastic stresses, the temperature oscillations may result in a transformation of the electromagnetic energy into acoustic energy (TEMA). A peculiar feature of the TEMA is that this mechanism is linear in spite of the fact that the dominating role in it is played by temperature oscillations. The linearity of the mechanism is manifested in the coincidence of the frequencies of electromagnetic and acoustic waves.

Investigations of the thermoelastic mechanism of TEMA require a knowledge of the limiting frequency ω at which this mechanism is operative. At a first glance, it seems that the thermoelastic mechanism cannot be extended to the anomalous skin-effect region in which $l \gg \delta$ (l is the electron mean free path and δ is the skin depth). However, this is not true. The temperature oscillations are restricted by the natural condition $\omega\tau \ll 1$, where $\tau = l/v$ is the relaxation time for quasiparticles responsible for heat transport, since temperature is a macroscopic quasiequilibrium concept. It was mentioned in Ref. 5 that temperature oscillations are usually not associated with the anomalous skin effect since the temperature skin-depth $\delta_T \sim l/\sqrt{\omega\tau} \gg l$ for $\omega\tau \ll 1$. However, temperature oscillations may emerge in a metal under the conditions of the anomalous electrodynamic skin effect ($l \gg \delta, \omega\tau \ll 1$).

If

$$\delta \ll l \ll \delta_T, \quad (1)$$

the source connected with the thermo-emf cannot be treated as a bulk source and must be replaced by an effective boundary condition (cf. Ref. 6).

If $q^e(z)$ is the density of the thermal flux transported by electrons dislodged from the equilibrium state by an electromagnetic wave of frequency ω (z is the coordinate along the normal to the surface), the temperature distribution for $z \gg l \gg \delta$ is defined (see Ref. 4) by the following asymptotic formulas:

$$\Theta(z) \approx \frac{e^{ik_T z}}{\kappa_{zz}} \begin{cases} \int_0^\infty q_z^e(z') dz', \\ -ik_T \int_0^\infty q_z^e(z') z' dz'. \end{cases} \quad (2)$$

$$(2a)$$

Formula (2) corresponds to the isothermal boundary, and (2a) to the adiabatic boundary, $T + \text{Re } \Theta(z, t)$ is the temperature in the metal ($|\Theta| \ll T$), the factor $e^{-i\omega t}$ is assumed to be present here and below, $k_T^2 = i\omega C/\kappa_{ik}$, $\text{Im } k_T > 0$, $\delta_T = (2\kappa_{zz}/\omega C)^{1/2}$, C is the specific heat of the metal, κ_{ik} the thermal conductivity tensor,¹⁾

$$q^e(z) = \frac{2}{(2\pi\hbar)^3} \int f_1(z, \mathbf{p}) \mathbf{v} (\varepsilon - \varepsilon_F) d^3 p, \quad (3)$$

where $f_1(z, \mathbf{p})$ is the nonequilibrium part of the electron distribution function, ε_F the Fermi energy, $\mathbf{v} = \partial \varepsilon(\mathbf{p})/\partial \mathbf{p}$ is the velocity of an electron with quasimomentum $\mathbf{p}$, and $\varepsilon = \varepsilon(\mathbf{p})$ its energy.

2. Thus, in order to solve the problem on the excitation of temperature oscillations under anomalous skin-effect conditions, we must know the function $f_1 = f_1(z, \mathbf{p})$, i.e., the solution of Boltzmann's kinetic equation.

In the τ -approximation, this equation can be written in the form

$$v_z \frac{\partial f_1}{\partial z} + \frac{f_1}{\tau} = - \frac{\partial f_F}{\partial \varepsilon} e v_i E_i(z). \quad (4)$$

Here, $E_i(z)$ are components of the electric field vector of a wave in the metal ($z > 0$), $f_F = f_F(\varepsilon)$ is the equilibrium Fermi function, and the derivative $-\partial f_F/\partial \varepsilon$ is close to the delta function $\delta(\varepsilon - \varepsilon_F)$ in the degenerate case (i.e., for $T \ll \varepsilon_F$). Formulation of the problem¹ requires the existence of an asymmetry, so that the normal incidence of an electromagnetic wave at the metal surface leads to the excitation of transverse as well as longitudinal component $E_z(z)$ of the electric field. The component $E_z(z)$ is determined from the condition

$$j_z(z) = 0, \quad (5)$$

where $\mathbf{j}$ is the electric current density:

$$\mathbf{j} = \frac{2e}{(2\pi\hbar)^3} \int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) \mathbf{v} f_1 d^3p. \quad (6)$$

Equation (5) is a corollary of the continuity equation $\text{div } \mathbf{j}=0$ and the absence of current through the metal surface.

3. Before proceeding further, let us calculate the temperature under the normal skin-effect conditions but in the terms that are convenient for comparison with the case of anomalous skin effect in which we are interested (cf. Refs. 1 and 2).

In the case of the normal skin effect, we should omit the diffusion term $v_z(\partial f_1/\partial z)$ in the kinetic equation. This gives

$$f_1 = -\frac{\partial f_F}{\partial \varepsilon} e \tau v_k E_k(z), \quad (7)$$

$$j_i = \sigma_{ik} E_k; \quad \sigma_{ik} = -\frac{2e^2}{(2\pi\hbar)^3} \int \frac{\partial f_F}{\partial \varepsilon} \varphi_{ik}(\varepsilon) d\varepsilon; \quad \varphi_{ik}(\varepsilon) = \oint \frac{\tau v_i v_k}{v} dS_\varepsilon. \quad (8)$$

Integration in the last formula is carried out over the constant energy surface $\varepsilon(\mathbf{p})=\varepsilon$, and dS_ε is an area element on this surface. It follows from (5) that

$$E_z(z) = -\frac{\sigma_{z\alpha}}{\sigma_{zz}} E_\alpha(z), \quad (9)$$

while the z -component of the thermal flux density has the form

$$q_z^e(z) = \frac{2e}{(2\pi\hbar)^3} \int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) (\varepsilon - \varepsilon_F) \left\{ \varphi_{z\alpha}(\varepsilon) - \varphi_{zz} \frac{\bar{\varphi}_{z\alpha}}{\bar{\varphi}_{zz}} \right\} d\varepsilon E_\alpha(z), \quad (10)$$

where

$$\bar{\varphi}_{ik} = \int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) \varphi_{ik}(\varepsilon) d\varepsilon. \quad (11)$$

Since

$$\int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) (\varepsilon - \varepsilon_F)^2 d\varepsilon = \frac{\pi^2}{6} T^2, \quad (12)$$

for $T \ll \varepsilon_F$, the first nonvanishing term in the expansion of $q_z^e(z)$ in powers of T/ε_F has the form

$$q_z^e(z) = \frac{\pi^2 e T^2}{3(2\pi\hbar)^3} \left\{ \varphi'_{z\alpha} - \varphi'_{zz} \frac{\partial_{z\alpha}}{\partial_{zz}} \right\} E_\alpha(z). \quad (13)$$

The prime indicates differentiation with respect to the energy ε , and all quantities $[\varphi_{ik}(\varepsilon), \varphi'_{ik}(\varepsilon)]$ are defined for $\varepsilon = \varepsilon_F$.

For definiteness, we assume that the incident wave is polarized along the x -axis (the x - and y -axes are chosen in such a way that the tensor $\sigma'_{\alpha\beta} = \sigma_{\alpha\beta} - (\sigma_{\alpha z} \sigma_{z\beta} / \sigma_{zz})$ is diagonal). In this case,

$$E_x(z) = E_x(0) e^{ik_E z}; \quad k_E^2 = \frac{4\pi i \omega}{c^2} \sigma'_{xx};$$

$$\frac{E_x}{H_y} = \frac{\omega}{ck_E}; \quad \text{Im } k_E > 0, \quad (14)$$

and

$$\int_0^\infty E_x(z) dz = \frac{E_x(0)}{ik_E} = \frac{i\omega}{ck_E^2} H_y(0); \quad \int_0^\infty z E_x(z) dz = -\frac{E_x(0)}{k_E^2} = -\frac{\omega}{ck_E^3} H_y(0). \quad (15)$$

Using formulas (2) and (2a), we obtain from (7), (14), and (15) as well as the Wiedemann–Franz law (see footnote 1)

$$\Theta(z) = \frac{e^{ik_T z} e T_c}{8\pi \sigma'_{xx}} \frac{\varphi'_{xz} \varphi_{zz} - \varphi'_{zz} \varphi_{xz}}{\varphi_{zz}} \left\{ \frac{1}{k_T/k_E} \right\} H_y(0). \quad (16)$$

The first row corresponds to the isothermal boundary [see (2)] and the second to the adiabatic boundary. Note that k_T/k_E is a real positive number of the order of δ_L/l , where $\delta_L \sim c/\Omega_L$ is the plasma (Langmuir) penetration depth, and $\Omega_L^2 = 4\pi n e^2/m^*$ is the square of the Plasma (Langmuir) frequency (m^* is the electron effective mass). We shall assume that $k_T/k_E \ll 1$ since we chose a surface heat source [see (2) and (2a)].

For estimates, we can assume that $\varphi'_{ik} \sim \varphi_{ik}/\varepsilon_F$. In this case, assuming that the anisotropy is of the order of unity, we obtain

$$\frac{\Theta(z)}{T} \sim e^{ik_T z} \frac{\omega_c}{\Omega_L} \frac{\delta_L}{l} \frac{c}{v_F} \left\{ \frac{1}{k_T/k_E} \right\}; \quad \omega_c = \frac{e H_y(0)}{m^* c}, \quad (17)$$

where v_F is the Fermi velocity. Here we have put $\varepsilon_F = m^* v_F^2/2$. Naturally, the temperature oscillations have a very small amplitude. The thermoelectric mechanism TEMA is perhaps the only possible means of detecting temperature oscillations (see Refs. 2 and 6).

It should be recalled that formulas (6)–(17) have been derived under the assumption that conditions corresponding to the normal skin effect prevail, i.e., $l \ll \delta$. Hence formula (16) and the estimate (17) are valid if the following inequalities are satisfied:

$$l \ll \delta \ll \delta_T, \quad T \ll \varepsilon_F. \quad (18)$$

4. Let us now turn to the case of the anomalous skin effect.

Strictly speaking, the solution of the kinetic equation cannot be obtained without taking into account the boundary conditions describing the interaction of conduction electrons with the sample surface. However, considering that the electrodynamic characteristics depend weakly on the boundary conditions in the case of extremely anomalous skin effect (see Ref. 7), the estimate for the effect can be obtained by using the following technique: it is assumed that the material equation connecting the electric field strength $\mathbf{E}$ and the current density $\mathbf{j}$ is the same as in unbounded space, and the solution of Maxwell's equations is sought in the class of discontinuous functions (tangential components of the vector $\mathbf{E}$ have a kink at the boundary $z=0$). This enables us to solve the problem on the reflection of an electromagnetic wave at the metal surface by calculating the tensor of impedances $\xi_{\alpha\beta}$ in the metal half-space:

$$E_\alpha = \xi_{\alpha\beta} [\mathbf{H}\mathbf{n}]_\beta; \quad \alpha\beta = x, y; \quad \mathbf{n} = (0, 0, 1). \quad (19)$$

In order to derive the material equation, we use the Fourier method and replace the functions depending on the coordinate z by their Fourier transforms:

$$E_i(k) = \int_{-\infty}^{\infty} E_i(z) e^{ikz} dz, \\ E_i(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E_i(k) e^{-ikz} dk \quad (20)$$

and so on. Note that the functions and their Fourier transforms are denoted by the same letters.

From (4)–(6), we obtain

$$f_1(k) = -\frac{\partial f_F}{\partial \varepsilon} e\tau \frac{v_i v_k}{1 + ikv_z \tau}; \\ \sigma_{ik}(k) = \frac{2e^2}{(2\pi\hbar)^3} \bar{\varphi}_{ik}(k); \\ j_i(k) = \sigma_{ik}(k) E_k(k); \\ \bar{\varphi}_{ik}(k) = \int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) \varphi_{ik}(k, \varepsilon) d\varepsilon; \\ \varphi_{ik}(k, \varepsilon) = \oint_{(\varepsilon)} \frac{dS}{v} \frac{\tau v_i v_n}{1 + (\tau k v_z)^2}; \\ E_z(k) = -\frac{\sigma_{z\beta}(k)}{\sigma_{zz}(k)} E_\beta(k); \\ j_\alpha(k) = \sigma'_{\alpha\beta}(k) E_\beta(k); \\ \sigma'_{\alpha\beta}(k) = \sigma_{\alpha\beta}(k) - \frac{\sigma_{\alpha z}(k) \sigma_{z\beta}(k)}{\sigma_{zz}(k)}. \quad (21)$$

These formulas (21) were written under the assumption that the constant energy surfaces have a center of inversion (because of this reason, the expressions do not contain the part of the distribution function f_1 odd in v_z). Note that all components of the conductivity tensor are even functions of the wave vector [$\sigma_{ik}(-k) = \sigma_{ik}(k)$].

In complete analogy with the above analysis, we can write

$$q_z^e(k) = \frac{2e}{(2\pi\hbar)^3} \int \left(-\frac{\partial f_F}{\partial \varepsilon} \right) (\varepsilon - \varepsilon_F) \left\{ \varphi_{z\alpha}(\varepsilon, k) - \varphi_{zz}(\varepsilon, k) \frac{\bar{\varphi}_{z\alpha}(k)}{\varphi_{zz}(k)} \right\} d\varepsilon E_\alpha(k) \quad (22)$$

or

$$q_z^e(k) = \frac{e\pi^2 T^2}{3(2\pi\hbar)^3} \left\{ \varphi'_{z\alpha}(k) - \varphi'_{zz}(k) \frac{\varphi_{z\alpha}(k)}{\varphi_{zz}(k)} \right\} E_\alpha(k). \quad (23)$$

As before, the prime indicates differentiation with respect to the energy ε , and all quantities [$\varphi_{ik}(\varepsilon, k)$; $\varphi'_{ik}(\varepsilon, k)$] are taken at $\varepsilon = \varepsilon_F$.

The expression (A1) presented in the Appendix describes the specific form of formula (23) for the case of an ellipsoidal energy-momentum relation for conduction electrons.

In order to determine $E_\alpha(z)$, we write Maxwell's equations:

$$\text{curl } \mathbf{H} = \frac{4\pi}{c} \mathbf{j}; \quad \text{curl } \mathbf{E} = \frac{i\omega}{c} \mathbf{H}. \quad (24)$$

Eliminating the magnetic field, we obtain

$$\frac{d^2 E_\alpha(z)}{dz^2} + \frac{4\pi i \omega}{c^2} j_\alpha(z) = 0. \quad (25)$$

To use Fourier's method, we continue the functions $E_\alpha(z)$ and $j_\alpha(z)$ to the halfspace $z < 0$ evenly: $E_\alpha(-z) = E_\alpha(z)$ and $j_\alpha(-z) = j_\alpha(z)$. The possibility of such a continuation does not contradict formulas (21). We direct the x - and y -axes along the principal axes of the tensor $\sigma_{\alpha\beta}(k)$ (for simplicity, we assume that the directions of the axes are independent of the magnitude k of the wave vector). Taking into account the fact that $E_\alpha(z)$ are even while $H_\alpha(z)$ are odd, we obtain from (21), (24), and (25)

$$E_{(y)}^x = \pm \frac{2i\omega}{c} \frac{H_{(x)}^{(0)}}{(4\pi i \omega / c^2) \sigma_{xx}^{(y)}(k) - k^2}; \\ H_\alpha(0) = H_\alpha(z=0). \quad (26)$$

This expression leads to the following principal values of impedance in accordance with the definition (19):

$$\xi_{(yy)}^{xx} = \pm \frac{2i\omega}{c^2} \int_0^\infty \frac{dk}{(4\pi i \omega / c^2) \sigma_{xx}^{(y)}(k) - k^2}. \quad (27)$$

The last formula has been presented only in order to complete the pattern.

Formulas (2), (20), (23), and (26) allow us to evaluate $\Theta(z)$. However, there is no need for calculations in the case of isothermal boundary. The temperature distribution is defined by the first rows of formulas (16) and (17) irrespective of the ratio l/δ . Indeed, since $E_\alpha(z)$ are even functions of the coordinate z and $\sigma_{ik}(k)$ are even functions of the wave vector $\mathbf{k}$, we obtain

$$\int_0^\infty q_z^e(z) dz = \frac{1}{2} \int_{-\infty}^\infty q_z^e(z) dz = \frac{1}{2} q_z^e(k) \Big|_{k=0}.$$

This equality is a corollary of the definition (20). For $k=0$, however, the values of all Fourier components of the kinetic coefficients coincide with their macroscopic values [cf. (21) and (17)]. This proves our statement.

(This statement may seem astonishing since it is well known^{7,8} that the penetration depth for the anomalous skin effect $\delta_{an} \sim (\delta^2 l)^{1/3} \neq \delta$, which should make the amplitude Θ of the thermal wave a function of the ratio δ/l . Indeed, if we are calculating the penetration depth of any quantity that does not depend exponentially on the coordinate, it is found that the penetration depth depends significantly on the way in which it is defined. The quantity δ_{an} is defined from the real dependence of $E_x(z)$ on the coordinate (see Ref. 8, p. 445).

This quantity has the meaning of the distance over which $E_x(z)$ changes (attenuates) significantly. If the penetration depth δ is defined as the ratio

$$\bar{\delta} = \frac{1}{E_x(z=0)} \int_0^\infty E_x(z) dz,$$

we obtain by using the relations (26) and (20)

$$\delta = \frac{c}{4\pi\sigma\xi} \sim \frac{\delta^{4/3}}{l^{1/3}}; \quad \delta = \frac{c}{2\pi\sigma\omega}.$$

($\xi = \xi_{xx}$, the anisotropy is neglected). If we had expressed Θ in terms of $E_x(0)$ instead of $H_y(0)$, the amplitudes of the thermal wave for normal and anomalous skin waves would differ in view of the dependence of ξ on l/δ .)

The situation is more complicated for an adiabatic boundary. An accurate estimate of the integral in formula (2a) is difficult since the dependences $\varphi_{ik}(k)$ and $\varphi'_{ik}(k)$ are determined to a considerable extent by the specific form of the dispersion relation for conduction electrons. To a very rough approximation, it can be assumed that the electron heat flux attenuates over a distance $\sim \delta_{an} \sim (\delta^2 l)^{1/3}$, where δ is the normal skin depth at a frequency ω (see Ref. 8). It can be seen from formula (2a) that, in the present case (adiabatic limit, anomalous skin effect), the temperature oscillation amplitude is smaller than the amplitude of oscillations in the isothermal limit by a factor of δ_T/δ_{an} .

Qualitative considerations presented in Ref. 4 explain such a ratio of the amplitudes of temperature oscillations. The amplitude of an acoustic wave excited in a metal due to thermoelastic stresses is also estimated there. Hence we shall not discuss this question in detail here.

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APPENDIX

Formula (23) can be presented in a more specific form by assuming that the Fermi surface is an ellipsoid (uniaxial for simplicity), whose axis is turned through an angle χ to the normal to the surface of the crystal. Let $m_{\parallel}$ and $m_{\perp}$ be the

effective masses of the conduction electrons. In this case, the expression in braces in formula (23) can be presented in the form

$$\begin{aligned} \Phi_{zx} &\equiv \varphi'_{zx}(k) - \varphi'_{zz}(k) \frac{\varphi_{zx}(k)}{\varphi_{zz}(k)} \\ &= - \frac{4\pi\sqrt{2\varepsilon_F m_{\parallel}^2 m_{\perp}}}{(\cos^2 \chi + \mu \sin^2 \chi)^2} \frac{\sin \chi \cos \chi}{k^2 \tau} \\ &\quad \times \left(1 - \frac{\arctan u}{u} \right) \frac{d\mu}{d\varepsilon} \Big|_{\varepsilon=\varepsilon_F}, \end{aligned} \quad (A1)$$

where

$$u = m_{\perp}/m_{\parallel},$$

and

$$u = k\tau\sqrt{2\varepsilon_F/m_{\perp}(\cos^2 \chi + \mu \sin^2 \chi)}.$$

For $m_{\parallel} = m_{\perp}$, the ratio $\mu \equiv 1$ and $d\mu/d\varepsilon = 0$, as expected. It should be emphasized that $\Phi_{zx} \equiv 0$ if the effective masses are constant. Essentially, this means that the effect under consideration is observed only beyond the quadratic spectrum of the conduction electrons. Of course, this conclusion is based on formulas obtained with the help of the kinetic equation in the τ -approximation, and is found to be valid in the case of specular reflection of electrons at the sample boundary.

¹We shall be interested in the region of low temperatures where electrons are mainly responsible for the heat transport, while scattering at the impurities (dislocations) is the main mechanism of dissipation. In this case, C is the electron specific heat ($C \propto T$) and $\kappa_{ik} = (\pi^2 T/3e^2)\sigma_{ik}$, where σ_{ik} is the electrical conductivity tensor $\sigma \approx ne^2 l/p_F$, n being the electron density and p_F the Fermi momentum.

¹M. I. Kaganov and V. M. Tsukernik, Zh. Éksp. Teor. Fiz. **35**, 474 (1958) [Sov. Phys. JETP **8**, 32 (1959)].

²M. I. Kaganov, Zh. Éksp. Teor. Fiz. **98**, 1828 (1990) [Sov. Phys. JETP **71**, 1028 (1990)].

³M. I. Kaganov and F. M. Maallavi, Fiz. Nizk. Temp. **18**, 737 (1992) [Sov. J. Low Temp. Phys. **18**, 521 (1992)].

⁴A. N. Vasil'ev, M. I. Kaganov and F. M. Maallavi, Usp. Fiz. Nauk **163**, 81 (1993) [Phys.-Usp. **36**, 81 (1993)].

⁵A. N. Vasil'ev, Yu. P. Gaidukov, M. I. Kaganov, and E. G. Kruglikov, Zh. Éksp. Teor. Fiz. **101**, 671 (1992) [Sov. Phys. JETP **74**, 357 (1992)].

⁶A. N. Vasil'ev and M. I. Kaganov, Usp. Fiz. Nauk **163**, 67 (1993) [Phys.-Usp. **36**, 392 (1993)].

⁷G. E. Renter and E. H. Sondheimer, Proc. Roy. Soc. **A195**, 336 (1948).

⁸E. M. Lifshitz and L. P. Pitaevskii, *Physical Kinetics*, Elsevier, 1981.

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